Invention Disclosure Details

1. Please provide a suitable title for your invention.

A) Drainage Rides

2. Please elaborate the problem being solved by your invention. Please specify the actual problem in the existing product or process.

A) Problem Statement:

1. Generally, most of the household waste and industrial waste gets disposed of into drainage systems.
2. This disposal includes solid waste too which gets accumulated in drainages and is referred to as debris generally.
3. Proper management of this type of waste in drainages is important to prevent blockages.
4. Many rural areas do not have an adequate amount of garbage bins on the roads for people to dispose of their waste. So people throw the garbage either on the vacant land or on the roadside.
5. Usually workers from municipalities clean the drainages and the trash on the road which involves direct contact with the waste which is not hygienic.
6. This may lead to infections, allergies and health problems and as the waste continuously accumulates in drainages day by day, workers need to continuously clean the drainages which only increases the exposure to pathogens, toxic gases from decomposing organic matter and chemicals from industries which leads to health hazards.
7. Using IoT-integrated automated machines to collect the trash on the roadside and clean waste in the drainages can help reduce human contact with the waste to a great extent which in turn decreases the disease-causing pathogens transmission.
8. These machines are embedded with different sensors to detect obstacles to avoid them and to detect the waste materials to collect them into trash bags or bins and they can work continuously. This improves hygiene and decreases health hazards.

Drawbacks in existing products or processes:

1. Most of the existing products need to be under the supervision of a human continuously like controlling the machine with a remote control for directing the path, collection of waste etc. And there are not many models using sensors for detection. Even if there are products that need not be controlled continuously and are sensor-embedded, they only concentrate only one aspect which is either the drainage cleanup or collecting the waste on the road.[1]
2. Sensors of the existing products that operate in the drainage might be damaged when there is an overflooding of drainage water due to various reasons.[5]

3. Please detail the solution being provided by the invention to overcome the problem. Please provide a schematic diagram and its description if required.

A) Recognizing the importance of these challenges, we attempt to come up with a novel design involving two separate machines. The two distinct machines that make up the

prototype design each have unique characteristics and purposes. It is designed to traverse roads and efficiently lift drainage waste from the drainage system.

Out of the two machines used in the design, one of the machines traverses on the road and the other machine is partially submerged in water. Both of these machines are connected through an extensible rod.

Case 1:

Functionality:

1. The machine partially submerged in water is equipped with a rotator consisting of three blades in a corner shovel-based structure. The machine is propelled forward inside the water with the assistance of motors and these three rotors rotate while being partially submerged in the water, collecting the garbage only. However, they drain the water and throw the garbage into a slanting rod-like structure that is attached to this machine and is also connected to the machine on land, where it is finally dumped into the bucket-like structure that is a part of the machine on land.

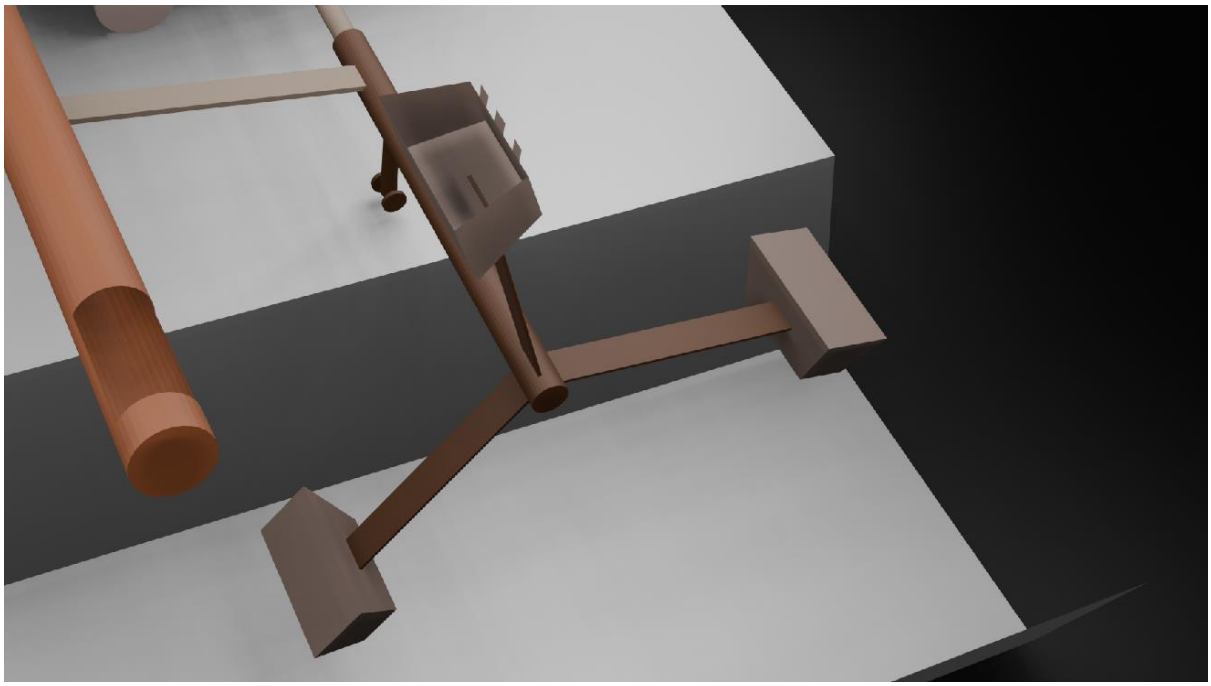

Fig:1

2.. The machine is powered by a voltage power supply and has motors that help it travel on the road. This machine consists of sensors that help it with the identification of obstacles and change its path when it detects heavy obstacles and uses the extensible rod to extend the distance between two machines so the machine on land can change its path while the machine that is semi-submerged in water will stay in its original path. This machine on land also uses a vacuum that uses suction to pull all the solid waste materials on the land beside open drainage and uses a filter to drain any liquid if it sucks any wet waste material. This machine also consists of a bucket-like structure for the storage of all the garbage collected before

disposal.

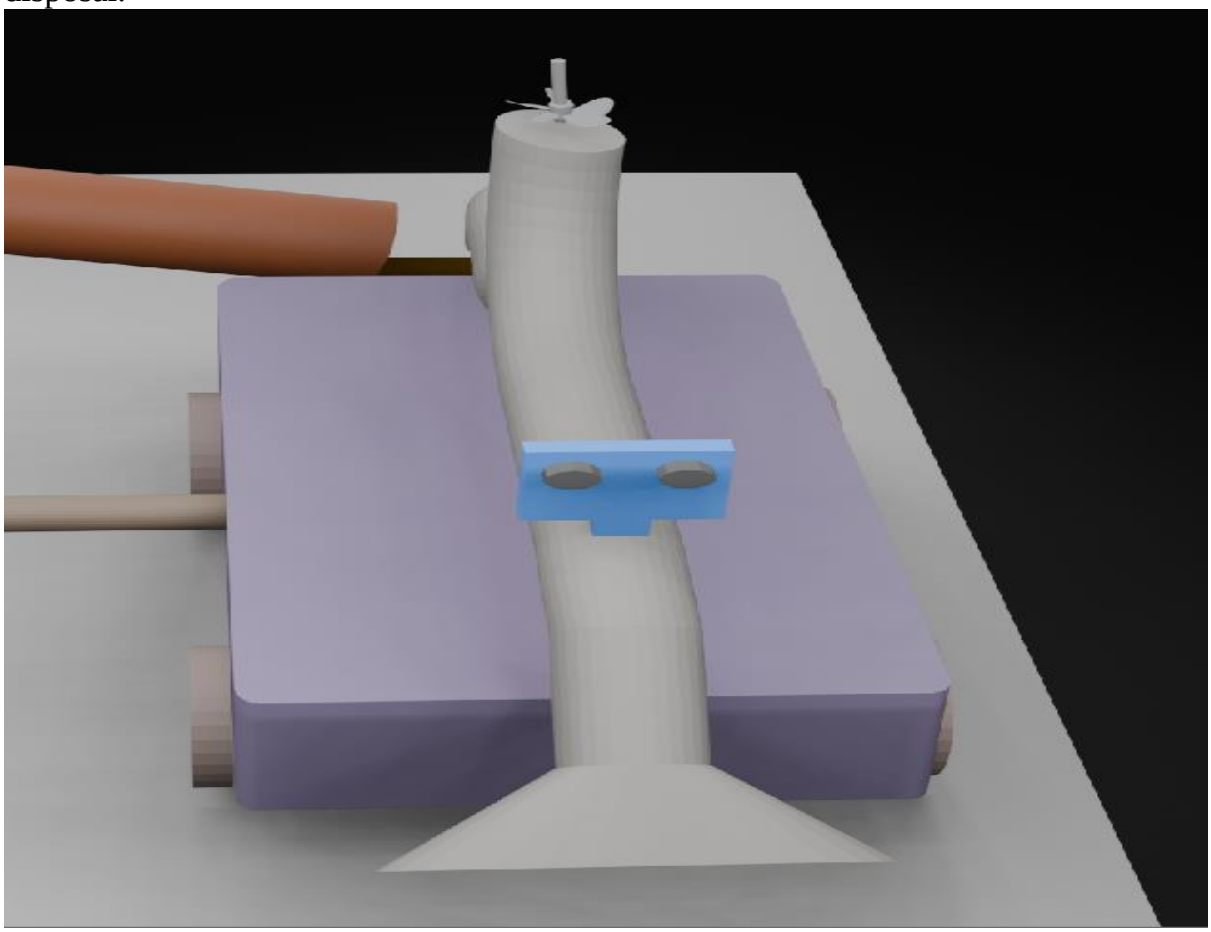

Fig:2

Case 2:

Operations: It works simultaneously cleaning the drainage along the premises. We can also separate the dust cleaning part for Other purposes of Cleaning. But we cannot use the Drainage cleaning part without the other.

Case 3:

User Interface

- Control Panel:
    - Simple controls for adjusting settings for each cleaning part.
- Indicators:
    - Alerts for maintenance needs or blockages.

Case 4:

Safety Features

- Automatic Shut-off:
    - Prevents overheating and damage to the machine.
- Emergency Stop Button:
    - Easily accessible for immediate shutdown.

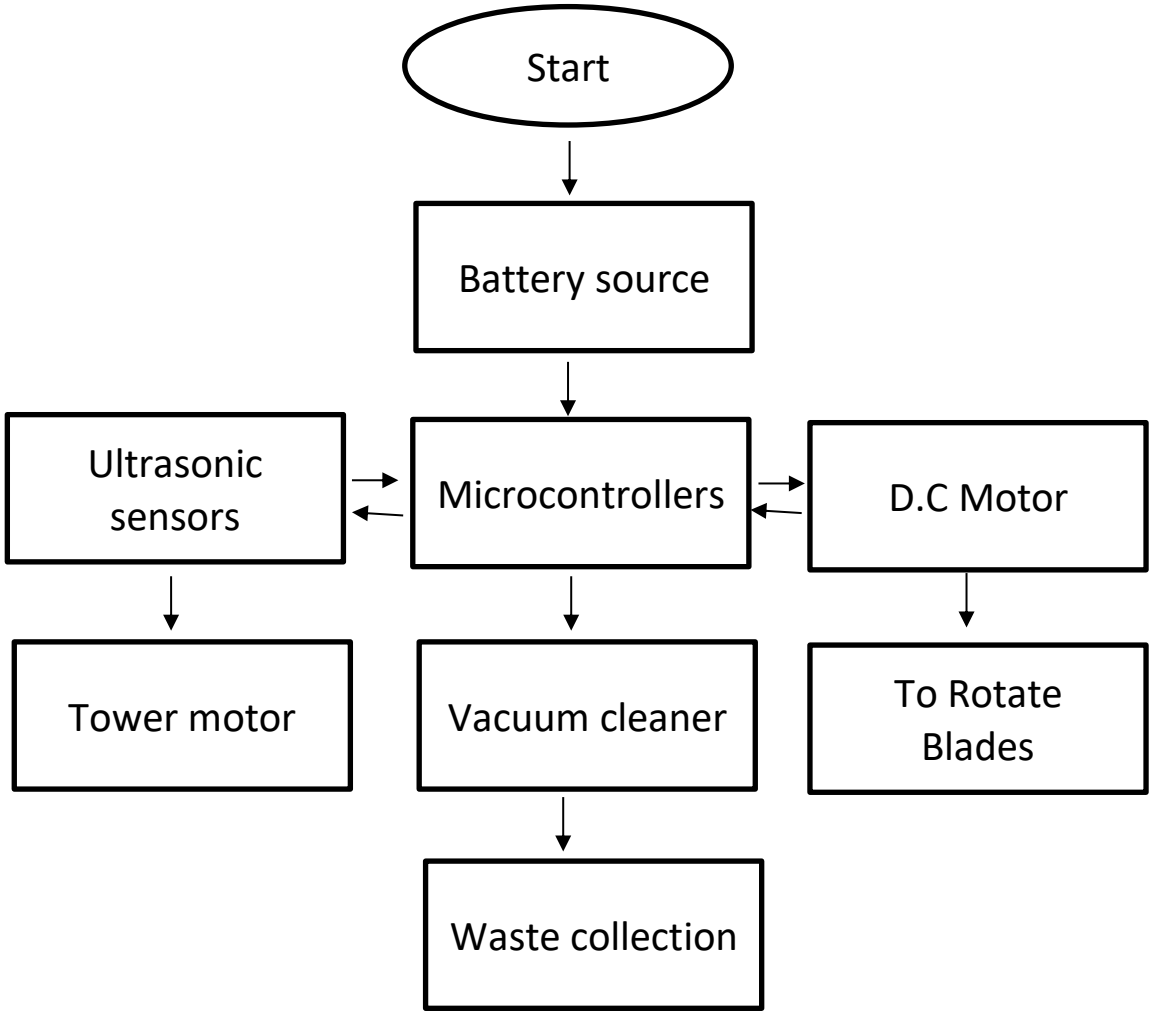
Start
Battery source
Ultrasonic sensors
Microcontrollers
D.C Motor
Tower motor
Vacuum cleaner
To Rotate Blades
Waste collection

Fig : 3

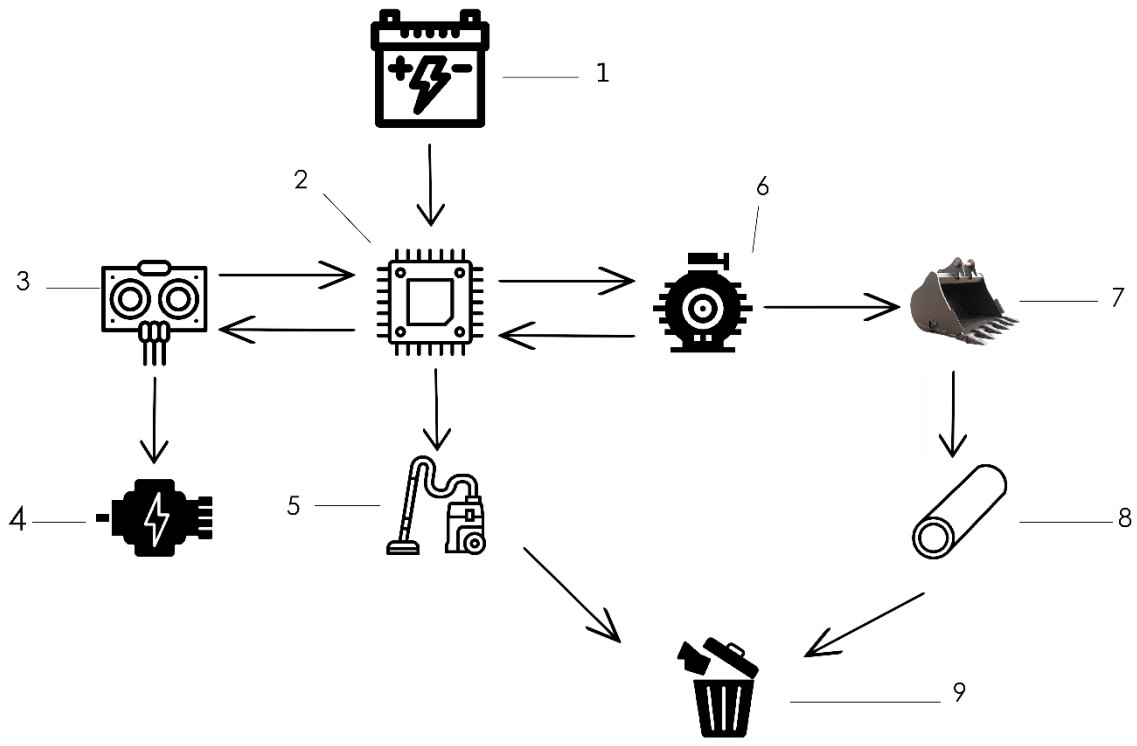

Fig:4

1. The battery source for the machine
2. It is the micro-controller of the machine. Which is Aurduino-UNO in our project. It co-ordinates with the L293N motor driver which helps from the fluctuations of current.
3. It is the ultrasonic sensor which detects the obstacle like big stones, human beigns or any obstacle. When it detects, it moves sidewards
4. These are the wheels to escape from the obstacles, to prevent the machine from the damage.
5. It is the vaccume cleaner, cleans up the road of dry waste. It throws the waste into the bin.
6. It is the HP-Motor which is responsible for the rotation of blades to lift the waste from the drainage.
7. The shovel which is in the shape of jcb bucket which can easily lifts the waste from the drainage and throws into a pipe. TO prevent the falling waste again into the drainage a plastic sheet is attached to the shovel such that it directly falls into it.
8. The lifted waste falls into this pip, which is fixed slanted such that the waste slides down easily to the bin.
9. The dustbin which collects the waste form the vaccum cleaner and pipe wchich are dry and wet wastes.

4. Please provide us the already available solution of the problem? Please provide us any information with you on the same.

A) Drainage Cleaning with arm robot: The proposed system aims to automate drainage cleaning by monitoring water levels without manual intervention. It includes a water level indicator, transmitter, receiver, and a robotic arm. The water level indicators are installed in each street drainage to monitor three levels of water: overflow, medium flow, and underflow. When the water reaches a certain level, the indicator triggers a transmitter, which sends messages regarding the water's status. The transmitter encodes these signals and transmits data via RF to a receiver located on street poles. The receiver decodes the signals and uses a GSM modem to alert maintenance workers via SMS. Additionally, the robotic arm, equipped with grippers, grinders, and suction tools, performs the cleaning tasks automatically, enhancing the efficiency of the drainage system.[5]

Smart Drainage Cleaning System: The project aims to automate drainage cleaning using water level sensors, pressurized motors, and an IoT module. It consists of sensors to detect water levels, a Raspberry Pi module, an Arduino Uno, an IoT module, and a pressurized motor. The sensors identify three levels of water flow based on clogs. The first level is cleared automatically using a pressurized pump. The second level can be cleared using auto mode with increased pressure or chemicals, or manual mode. The third level, above 300, requires manual intervention. The system utilizes ATMEGA328P, Raspberry Pi 3, and pressure booster pumps to achieve its objectives.[4]

Automatic Drainage Cleaning: Numerous projects are based on the conveyor belt system. This system comprises a conveyor belt and baskets designed for the collection of waste. Control of the system can be executed manually by operators or automated through the use of Internet of Things (IoT) technology. In the IoT implementation, devices such as Raspberry Pi or Arduino can be utilized, along with microcontroller-based monitoring to assess the levels of drainage waste.[6][7]

5. Please provide you with how your solution is unique and different as compared with other available/known solutions to the same/similar problems?

A) We have employed sensors previously used by others for waste extraction from drainages. Our approach, however, introduces a unique solution aimed at maintaining hygiene across open drainage systems and their surroundings. Our system features rotator mechanisms designed to lift waste efficiently. Additionally, we employ sensor-equipped machines connected to extendable rods, facilitating ground-level movement. These machines effectively clean up the surroundings, ensuring thorough waste removal. Unlike conventional methods that often float on drainage surfaces, our system operates from the road's edge, ensuring effective waste management without disrupting the drainage flow and it cleans both the open drainages and its surroundings simultaneously.

6. Please provide us with the advantages of your solution. Please Compare the advantages in view of the existing/conventional solutions uncovered during the search under point. You may make a table or figures. Further, you can provide us with the economic potential or commercial applications for the technology.

A) Advantages:

- This machine can operate continuously without fatigue.
- It ensures thorough cleaning.
- Machines can work faster than manual methods.
- Reduces human contact.
- Improves hygiene and sanitation.
- And both machines can work together or individually also (Coordinated operations).
- Cost effective and reduces labour costs.

Our Product	**Existing Product**
Time saving and operational flexibility.	Task optimisation but takes more time to complete both the tasks.
Can be cleaned both drainage and premises of it.	Only cleaning the drainage[2].
we use vacuum to clean up the waste on the road which is similar to household vacuum cleaner but on a larger scale to collect debris, litter, dried leaves etc.	using other methods such as robotic arms, mechanical sweepers etc.
Bucket Systems for lifting the waste.	Conveyor belt operation for lifting the waste[3].
Completely Sensors equipped and no need for manpower to control it.	Some need to be controlled remotely or by humans[1].

7. Please intimate if can think of any alternative way/solution of achieving the same result as your invention. Please note that the alternative way/solution that you are thinking of may or may not have the same advantages as offered by the invention.

A) No.

8. What are the novel aspects of your invention that need protection? Describe the technical difference between the conventional and available product/process and your solution.

A.

1. Vacuum is used in our land machine to suck the waste in the surroundings of the open drainage system.
2. This invention can help us clean both the drainage and the surroundings of the open drainage system simultaneously.

9. Any environmental issues confronted while developing or implementing the invention?

A) No.

10. Please enlist all innovative features of your invention that you think are making the invention more sophisticated.

A.

1. This invention helps us to clean both the drainage and the surroundings of the open drainage system simultaneously.
2. It helps us in the efficient utilization of time and energy as it eases the process of cleaning the open drainage systems and also helps in reducing human contact which decreases the number of people prone to waterborne diseases and harmful chemicals while cleaning the drainages.
3. The machine working on the land has a vacuum as a part of it which helps it to collect all the solid waste from the surroundings of the open drainage system.
4. This machine consists of both components working on the land and another semi-submerged in the drainage connected through an extensible rod which can be either used together or can be used as individual components also.

11. What are the utilities/applications of the invention? Enlist them.

A)

- This machine can be left on the roads which are margined with open drainages.
- This can be used in construction sites for constant cleaning.
- Can be used in residential areas.
- It is useful for collecting leaves and garbage like paper cups in the parks.

12. Is the invention already been implemented in any product or process?

A) No.

13. Has the invention been published or disclosed/discussed to anyone outside of your organization or any third party including abroad also (such as marketing

meetings, conferences, tradeshows, trade fairs, websites, social media, newspapers etc.)?

A) No.

14. Inventor details in the below format

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